

Exercise 4

Original Code generation: The `#pragma omp parallel for` directive only parallelizes the outermost loop (the `r` loop). Because the outer loop only has 5 iterations, a maximum of 5 threads can be assigned work. Whichever thread is assigned a specific `r` iteration will execute the entire inner `k` loop (all 10 iterations) sequentially on its own.

With `collapse(2)` generation: Adding `collapse(2)` instructs OpenMP to merge the two nested loops into a single, flattened iteration space of 50 iterations (5×10). These 50 independent iterations are then distributed across the available thread pool. This allows the program to utilize more than 5 threads, significantly improving workload distribution and load balancing.